PV101 Day 8 Student Print Packet

Compiled packet

Day_08_PV101_Final_Review_and_Assessment_2026-07-16

Contents follow in instructional order.

PV101 Day 8 Morning Quiz

Student copy | 10-15 minutes

Day_08_PV101_Final_Review_and_Assessment_2026-07-16

Name: _____ Date: _____

Answer briefly. Show work and units where needed.

1. A 24 V training load draws 2 A. What power does it use?

_____

_____

2. A 40 W device operates for 2 hours. What daily energy does it use?

_____

_____

3. Before testing continuity across a fuse, name the required energy state and meter function.

_____

_____

4. A wire label says polarity is correct. What evidence should verify the label?

_____

_____

5. Why write the expected value and unit before taking a measurement?

_____

_____

PV101 Day 8 Review Worksheet

Student copy | Use before the final

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Name: _____ Date: _____

Formula and units check

Write the power formula: _____

Calculate 24 W used for 3 h: _____ Wh

Convert 0.5 kW to watts: _____ W

In series, _____ adds. In parallel, _____ adds.

Verification chain

Put in order: evidence, symptom, conclusion, test.
Answer: _____

System path

Draw and label the basic PV-to-load energy path. Mark one fuse, one disconnect, and one meter point.

_____

_____

_____

Common misconceptions to correct

1. Wire color proves polarity. Correct / Incorrect: _____
Correction: _____

2. Continuity is tested on an energized circuit. Correct / Incorrect: _____
Correction: _____

3. Replacing a fuse proves the fault is repaired. Correct / Incorrect: _____
Correction: _____

4. Loads are the starting point for sizing. Correct / Incorrect: _____
Correction: _____

5. Cleanup is part of technical work. Correct / Incorrect: _____
Correction: _____

PV101 Day 8 Guided-Practice Guide

Student-facing

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Name: _____ Date: _____

Purpose

Provide the student-facing sequence for review, final assessment, reflection, and PV101 closeout.

Materials

- Readiness quiz
- Review worksheet
- Calculator and pencil
- Final assessment
- Reflection questions

Student sequence

- Complete the readiness quiz.
- Participate in cumulative review.
- Use lunch as a test-content break.
- Complete the final independently.
- Turn in the final before starting reflection.
- Submit reflection and leave the room reset.

Test rules

- One point per question.
- Show units on calculations.
- Ask only for clarification of directions.
- Do not discuss or photograph test content.

Cleanup

- Return calculators.
- Submit every test and scratch page.
- Upload/submit reflection as directed.
- Clear tables and wait for dismissal.

PV101 Day 8 Reflection Questions

Student copy

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Name: _____ Date: _____

1. What PV101 skill improved most for you?

_____

_____

2. Which safety or verification rule will you remember?

_____

_____

3. Which topic still needs practice?

_____

_____

4. Describe one time evidence changed your conclusion.

_____

_____

5. What does professional cleanup/documentation mean to you?

_____

_____

PV101 Final Assessment

Student copy | 25 questions | 25 points

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Name: _____ Date: _____ Score: _____ / 25

Directions: Work independently. Circle one answer for multiple choice. Show formulas and units for calculations. Write clearly for explanations. Each question is worth one point.

1. Photovoltaic (PV) equipment converts which resource directly into electricity?

- A. Moving air
- B. Light
- C. Flowing water
- D. Fuel vapor

2. Which application most clearly needs energy storage to operate when sunlight is unavailable?

- A. A direct daytime PV fan
- B. A grid-tied array with no backup requirement
- C. An off-grid cabin used at night
- D. A module demonstration with no load

3. Before changing wiring on a training PV system, the safest first action is to:

- A. Work faster
- B. Isolate energy sources and verify the state
- C. Trust the switch label
- D. Replace the fuse

4. A student is uncertain about polarity. What should happen?

- A. Connect briefly and watch
- B. Ask a classmate to guess
- C. Stop and verify with the correct meter setup
- D. Reverse both conductors

5. Which statement is correct?

- A. Voltage is electrical pressure and current is flow
- B. Current is measured in volts
- C. Power is measured in ohms
- D. Resistance creates energy

6. A 12 V fan draws 3 A. Calculate its power in watts. Show the formula.

_____

_____

_____

7. Peak sun hours are used mainly to estimate:

- A. Daily equivalent full-sun energy
- B. Wire color
- C. Battery terminal size
- D. AC frequency

8. What usually happens to PV output when a module is heavily shaded?

- A. Output improves
- B. Output is reduced
- C. AC voltage doubles
- D. Battery capacity increases

9. A PV module exposed to light should be treated as:

- A. Always de-energized
- B. A possible DC voltage source
- C. An AC-only device
- D. A fuse

10. Read this connection diagram: [Module 1] -- series -- [Module 2]. Which electrical quantity normally adds in this matched series connection?

_____

_____

_____

11. What is the main job of a charge controller in a battery-based PV system?

- A. Convert DC to AC
- B. Regulate charging to protect the battery
- C. Create sunlight
- D. Replace overcurrent protection

12. What is the main job of an inverter?

- A. Convert DC to AC
- B. Store energy
- C. Measure resistance
- D. Increase sun hours

13. Three 10 W lights operate 4 hours per day. Calculate their total daily energy use.

_____

_____

_____

14. What should be completed before selecting battery, array, or inverter size?

- A. A load assessment
- B. A wire-color chart
- C. A tool inventory only
- D. A final photograph

15. Why is a fuse placed close to a battery source?

- A. To increase voltage
- B. To protect as much conductor as practical
- C. To make the inverter quieter
- D. To eliminate shutdown steps

16. A long undersized conductor carrying current is most likely to cause:

- A. Lower voltage drop
- B. Voltage drop and heating
- C. More sun hours
- D. Increased battery capacity

17. Which is a reasonable advantage of a ground-mounted array?

- A. It always costs less
- B. It can be easier to access and orient
- C. It cannot be shaded
- D. It needs no structure

18. A roof area is shaded from 1:00-4:00 PM, while an open ground area has clear solar access. Circle the better conceptual location and write one reason.

_____

_____

_____

19. After observing a fault symptom, the best next step is to:

- A. Replace several parts
- B. Choose one test that can separate likely causes
- C. Bypass protection
- D. Trust the label

20. Explain why replacing a blown fuse is not enough to prove the system is repaired.

_____

_____

_____

_____

_____

21. In a permitting context, the AHJ is the organization or official that:

- A. Guarantees system output
- B. Sets the price of every module
- C. Enforces the applicable local requirements and approvals
- D. Replaces the serving utility

22. A current, documented rebate in a training example covers $500 of a $2,500 cost. Calculate the remaining cost before any other fees or credits.

_____

_____

_____

23. After completing a set of practice questions, explain why reviewing the missed questions is more useful than recording only the score.

_____

_____

_____

_____

_____

24. A cumulative practice question asks you to convert 0.75 kW to watts. Show the conversion.

_____

_____

_____

25. A small PV training system does not power its load. The panel is in light, the controller display is off, and a label says the battery fuse is good. Describe the safe first three actions/tests and the evidence you would document.

_____

_____

_____

_____

_____